

SILAS M. FALDE

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EDUCATION

University of Michigan School of Engineering
Senior B.E.S. in Data Science, GPA: 3.09/4.00

Ann Arbor, MI
August 2023 – December 2026

Relevant Courses: Machine Learning, Artificial Intelligence, Computer Vision, Computational Modelling of Complex Systems, Database Management, Regression Analysis, Probability and Statistics, Bayesian Data Analysis, Data Structures and Algorithms

EXPERIENCE

Ally Financial, Inc.

Intern, Artificial Intelligence & Advanced Analytics

Detroit, MI
May 2026 – August 2026

- **Prototyped large language model workflows** to automate internal tasks and improve efficiency and safety.
- **Built Python NLP pipelines** to evaluate PII masking and redaction model performance for sensitive company documents.
- **Enhanced** a data dictionary migration tool by adding **batch processing**, smarter metadata selection, and a streamlined interface to improve usability and efficiency.

University of Michigan Department of Epidemiology

Research Assistant – Computational Lead

Ann Arbor, MI
September 2025 – May 2026

- Led development of a Python pipeline to classify collective-violence narratives using prompt-based large language model inference and structured response processing.
- Built reusable preprocessing utilities for NVDRS-style text, demographic derivation, path management, and annotation-round data handling.
- Designed human annotation workflows and measured inter-annotator agreement with Cohen's kappa and Krippendorff's alpha to resolve disagreements and assess model performance.

Ally Financial, Inc.

Intern, Data Management

Detroit, MI
May 2025 – August 2025

- **Created** data dictionaries and lineage documentation in **Python** to improve data governance and compliance.
- **Added data quality checks** to existing data pipelines with **Python and SQL** to improve data integrity and reliability.
- **Developed** a **Power BI** data availability dashboard to monitor refresh rates and availability for critical data assets.

PROJECTS

Predicting Survival in the ICU

Project Developer

Ann Arbor, MI
January 2025 – March 2025

- **Developed** logistic-regression, convolutional, and recurrent-neural-network models in **Scikit-learn** and **PyTorch** to predict ICU mortality from clinical time-series and static features.
- **Built reproducible data loaders and preprocessing** for missing values, temporal observations, and engineered clinical features, with configuration-driven training and evaluation.
- **Evaluated models** with **AUROC**, visualization, and automated tests covering CNN, transformer, and output behavior.

Capuchin Audio Classification

Machine Learning Developer

Ann Arbor, MI
Course Project

- **Built an end-to-end PyTorch pipeline** to detect capuchin monkey calls in forest recordings by loading, resampling, and batching variable-length audio.
- **Converted recordings to padded spectrogram tensors** and trained a convolutional neural network with GPU support, checkpointing, and TensorBoard loss tracking.
- **Ran inference** over field recordings and exported per-recording call counts to CSV for downstream analysis.

Vaccine Uptake Prediction

Data Science Developer

Ann Arbor, MI
Course Project

- **Built a reproducible Python pipeline** to predict multilabel vaccine uptake from NHFS survey data, including preprocessing, fold creation, and model evaluation.
- **Compared regularized logistic regression, random forests, and Bayesian hierarchical models** using cross-validation, ROC AUC, Brier scores, and calibration plots.
- **Organized configuration, data utilities, model utilities, and evaluation code** into a modular source package with a notebook-driven workflow.

Image Classification and Transfer Learning

Computer Vision Developer

Ann Arbor, MI
Course Project

- **Built a complete Python/PyTorch image-classification workflow** with dataset preparation, CNN and Vision Transformer models, training scripts, and challenge inference.
- **Compared custom CNNs, transfer-learning source and target models, and Vision Transformers**, with validation AUROC reported around **0.97** for CNN experiments.
- **Added automated tests** for datasets, image standardization, CNNs, ViTs, and generated outputs to validate the training and evaluation pipeline.

Breast Cancer Detection

Machine Learning Developer

Ann Arbor, MI
Course Project

- **Investigated breast-cancer image classification** with Python notebooks covering exploratory data analysis, convolutional neural networks, EfficientNetV2, and Vision Transformer models.
- **Built image preprocessing, training, and evaluation experiments** with NumPy, pandas, Pillow, Matplotlib, and PyTorch-oriented workflows.
- **Compared transfer-learning approaches** and documented model behavior through visual analysis and experiment reports.

SKILLS

Programming: Python, C++, Bash, JavaScript, R, SQL

Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, spaCy, NLTK

Web: HTML, CSS, JavaScript, responsive design, static-site publishing

Systems: Linux, Docker Compose, systemd, CMake, Make, Git, CI/CD

Software: VS Code, Microsoft Office Suite, Google Workspace, Snowflake, Collibra, Alation, Manta